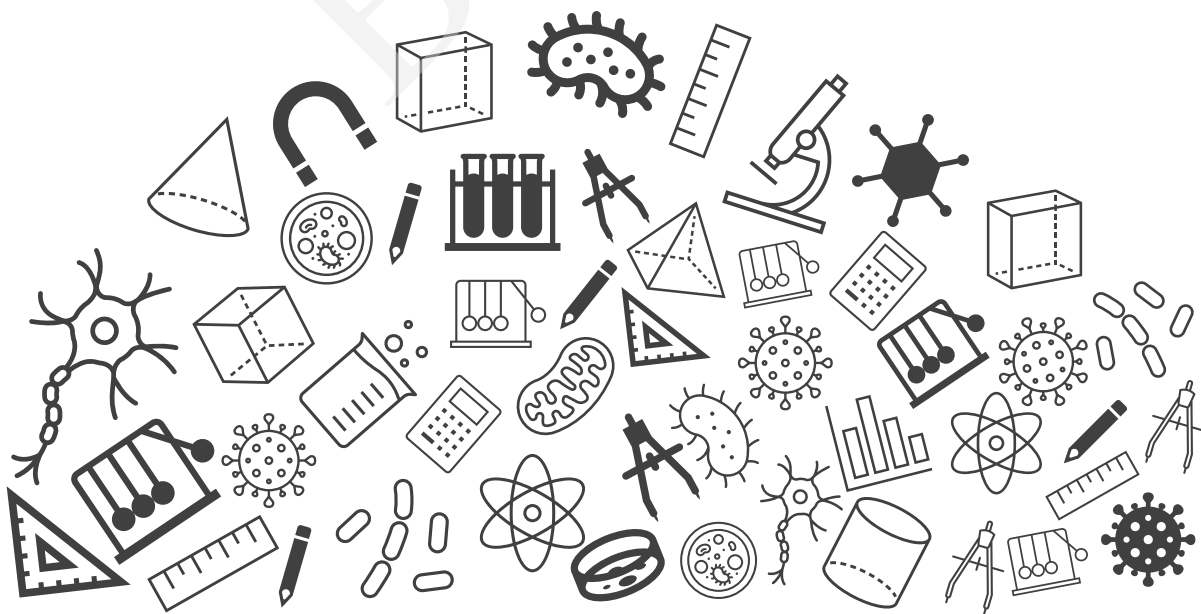




Science Chapter Notes





BYJU'S Classes

Chapter Notes

Is Matter around Us Pure?

Grade 09



Topics to Be Covered

1

1. Pure Substances

1.1 Elements

1.2 Compounds

2

2. Mixtures

2.1 Homogeneous

2.1.1 Solutions

2.2 Heterogeneous

2.2.1 Suspensions

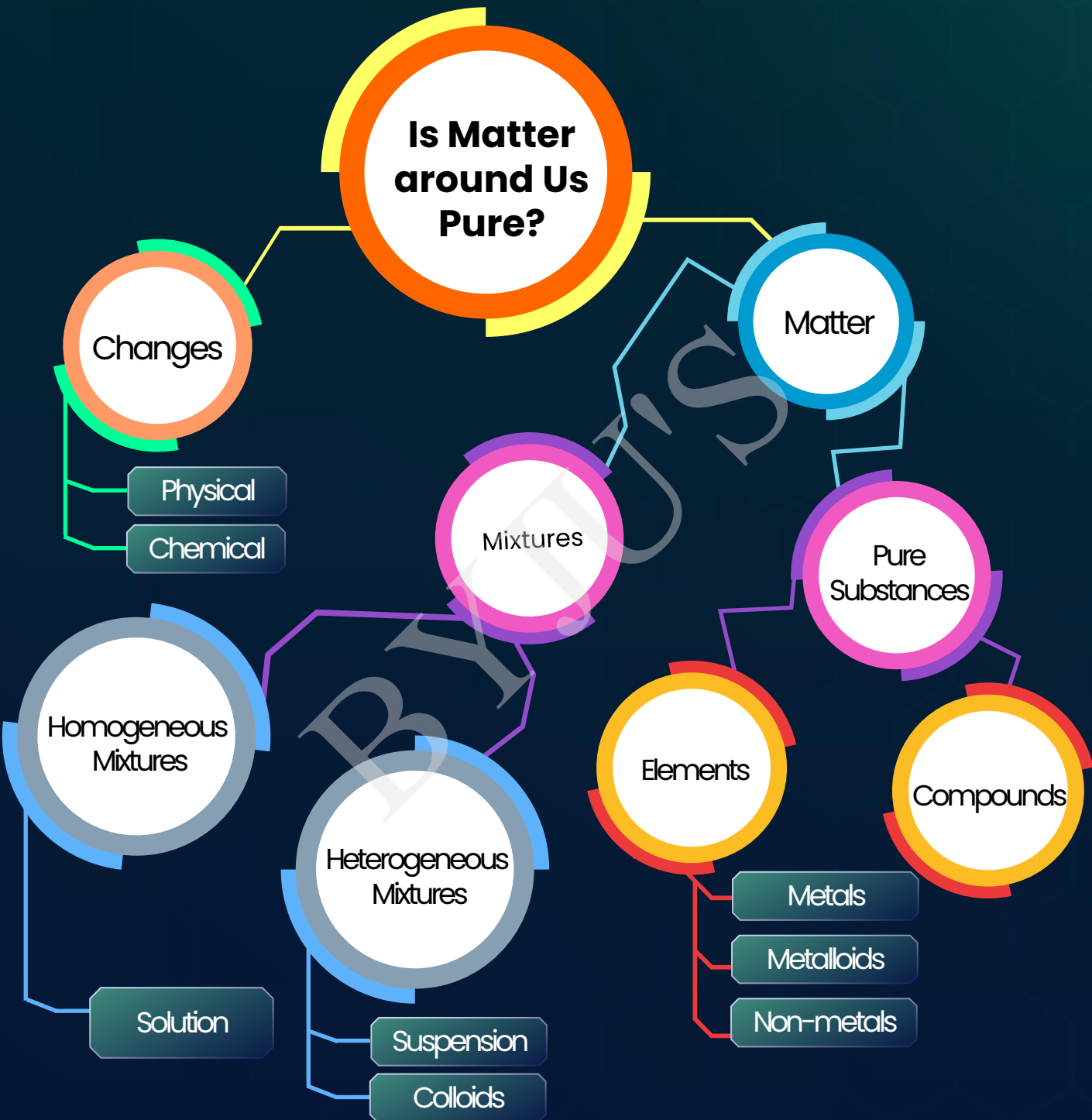
2.2.2 Colloids

3

2. Changes

4.1 Physical

4.2 Chemical



1. Pure Substances

A form of matter having an invariant chemical composition and properties that are constant throughout the sample.

Types of Pure Substances

1.1 Elements

- Pure fundamental substances.
- They cannot be broken down further by chemical reactions.
- **Example:** Iron, sulphur, and more.



1.2 Compounds

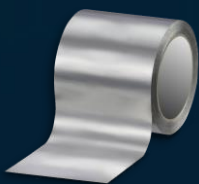
- Substances composed of two or more elements.
- They can be broken down further by chemical reactions.
- **Example:** Iron sulphide, water, and more



1. Pure Substances

Classification of Elements

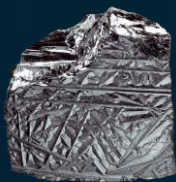
Metals



Eg: Aluminium

- Lustrous
- Malleable
- Ductile
- Sonorous
- Hard
- Good conductor

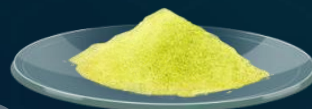
Metalloids



Eg: Silicon

Have intermediate properties between those of metals and non-metals.

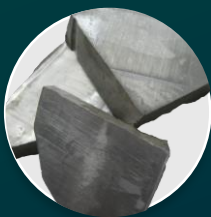
Non-metals



Eg: Sulphur

- Non-lustrous
- Non-malleable
- Non-ductile
- Non-sonorous
- Soft
- Poor conductor

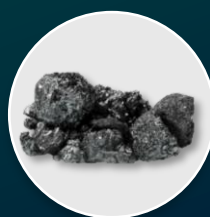
Exceptions



Sodium
Soft



Mercury
Liquid at room temperature



Iodine
Lustrous

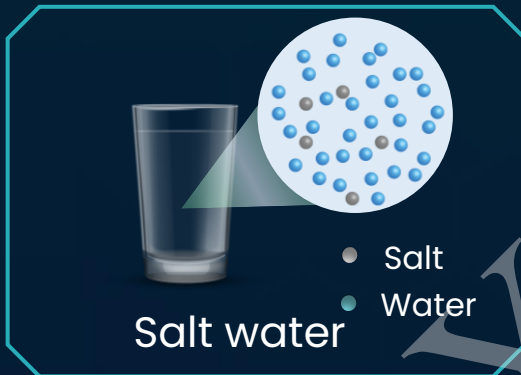


Graphite
Conductivity

2. Mixtures

Mixtures are formed by physical combination of two or more pure substances. A mixture has a variable composition. The constituents can be separated easily by physical methods.

Examples:



Types of Mixtures

Homogeneous

- Such mixtures will have uniform composition.

Heterogeneous

- Such mixtures will have non-uniform composition.

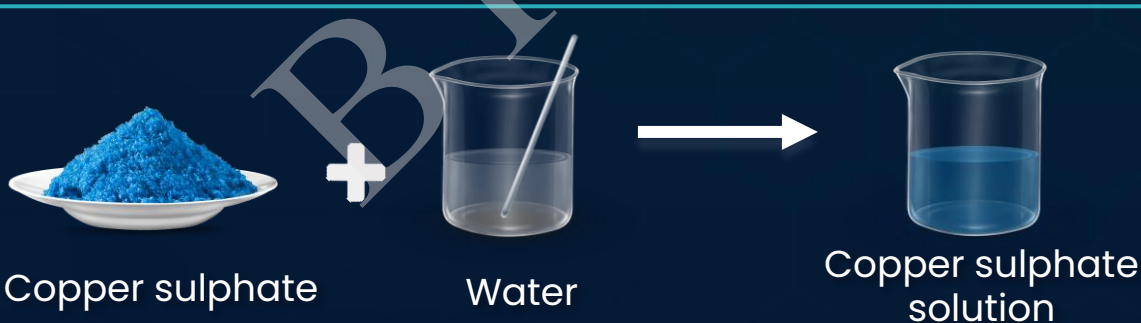
2. Mixtures

2.1 Homogeneous Mixtures

Homogeneous mixtures will have a uniform distribution of components. Components of homogeneous mixture cannot be distinguished.

2.1.1 Solution

A solution is a homogeneous mixture of two or more components in which the constituents are not visible to the naked eye (due to very small particle size, which is less than 1 nm).



- The components of a solution include the solute (minor quantity) and the solvent (major quantity).
- The solute particles cannot be separated from the mixture by the process of filtration, and they won't settle down also.

2. Mixtures



Tyndall effect: It is the phenomenon by which the path of the light becomes visible in a mixture due to scattering of light by the particles.

Due to small particle size, solutions do not show Tyndall effect

Types of Solution

1 Saturated

At any particular temperature, a solution that has dissolved as much solute as it is capable of dissolving, is said to be a saturated solution.

2 Unsaturated

If the amount of solute contained in a solution is less than the saturation level, it is called an unsaturated solution.

3 Supersaturated

A supersaturated solution is one in which more solute is dissolved than is necessary to make a saturated solution.

2. Mixtures

Concentration of a Solution

Concentration of a solution is the amount of solute present in a given amount of solution.



Dilute



Concentrated

Depending upon the amount of solute present in a solution, we can refer it as a dilute or concentrated solution.



Dilution: The process of decreasing the concentration of a solute in a solution that is usually done by adding more solvent in it.

2. Mixtures

Ways of Expressing the Concentration of a Solution

1

Mass by mass
percentage of a
solution

$$= \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

2

Mass by volume
percentage of a
solution

$$= \frac{\text{Mass of solute}}{\text{Volume of solution}} \times 100$$

3

Volume by
volume
percentage of a
solution

$$= \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

2. Mixtures

2.2 Heterogeneous Mixtures

2.2.1 Suspension

A suspension is defined as a **heterogeneous** mixture in which the solid particles are solids are dispersed in liquids.

The particles of suspension:

- Are easily visible and distinguishable.
- Can be separated by filtration.
- Settle down when left undisturbed.

Suspensions in our daily life



Sand in water



Tea leaves in water



Chalk powder in water

2. Mixtures

2.2.2 Colloids

Colloids: A heterogeneous mixture in which the particles are uniformly spread throughout the solution. Hence it appears to be homogeneous.

The particles of colloids:

- Is too small to be individually seen by naked eyes.
- Are big enough to scatter a beam of light passing through it and make its path visible.
- Will not settle down when left undisturbed.

Colloids in our everyday life



Milk

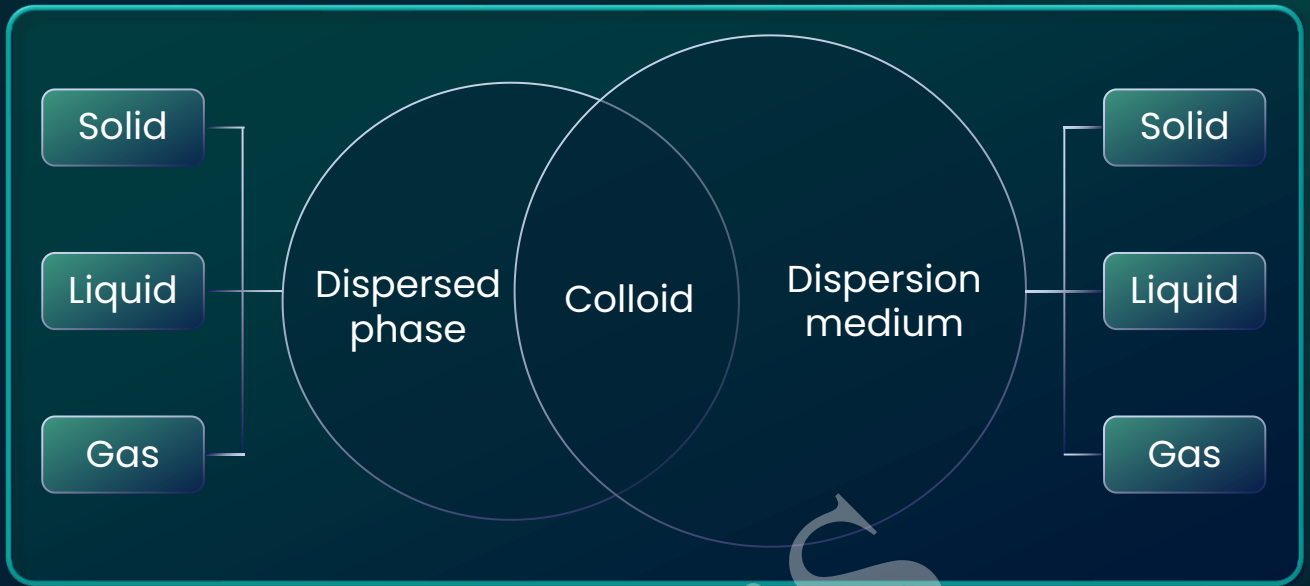


Ice cream



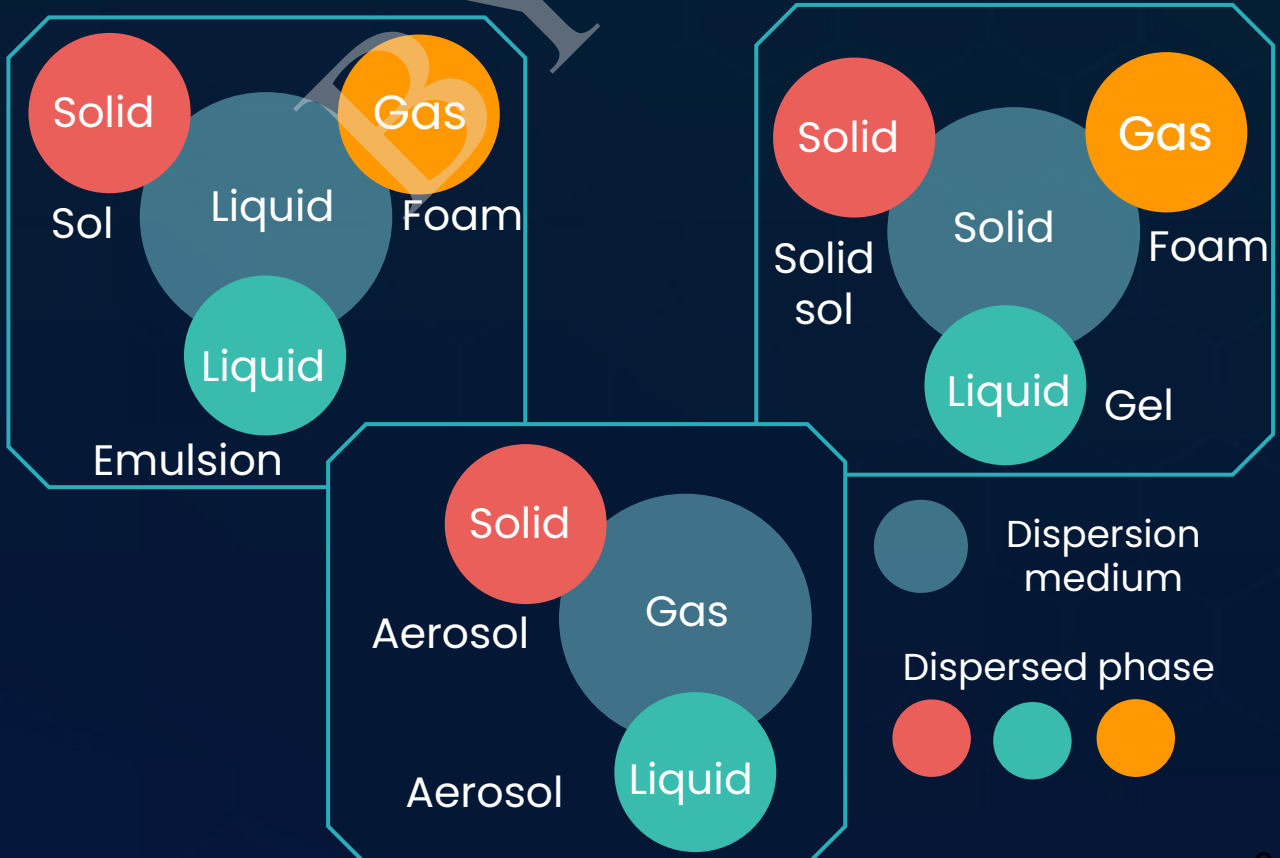
Paint

Components of a Colloid



Classification of Colloids

According to the state (solid, liquid, or gas) of the dispersing medium and the dispersed phase, colloids are classified into different types.



Comparison

	Solution	Colloid	Suspension
Particle size	Less than 1 nm	Between 1-1000 nm	More than 1000 nm
Tyndall Effect			

The table compares three types of mixtures: Solution, Colloid, and Suspension. For each mixture, it shows the particle size and whether it exhibits the Tyndall effect. Solutions have particles smaller than 1 nm and do not show the Tyndall effect. Colloids have particles between 1 and 1000 nm and do show the Tyndall effect. Suspensions have particles larger than 1000 nm and also show the Tyndall effect. Each entry includes a beaker diagram with a magnified view of the particles.

3. Changes

3.1 Physical Change

A physical change is a change in which the physical properties such as size, state, shape, appearance, and more of a substance alters without changing its chemical nature.

Mostly, physical changes tend to be reversible.

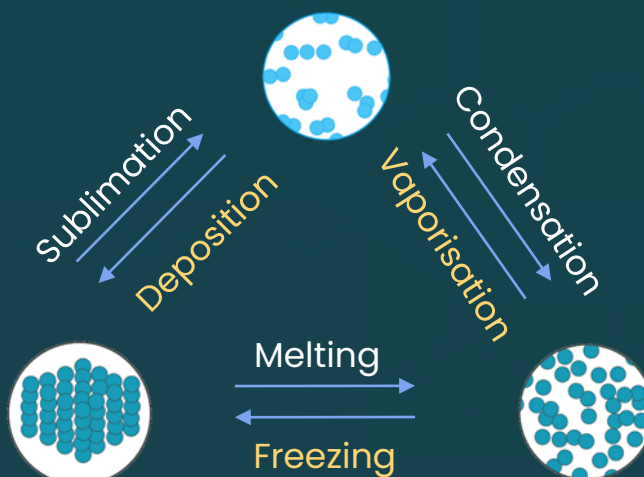


Melting ice



Tearing a paper

Interconversion of states of matter is a physical change:



3. Changes

3.2 Chemical Change

A chemical change is a change that brings change in the chemical properties, and we get new substances.

A chemical change is frequently harder to reverse than a physical change.



Burning of wood



Rotting of an apple



Formation of curd



During burning of a candle, both physical and chemical changes take place.

Burning wick (Chemical change)

Melting wax (Physical change)

